

Čech complexes for finite sets

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Abstract

Let $\mathcal{F}^m = \{A \subset [m]\}$, endowed with the symmetric difference metric and the Cardinality Reverse-Lexicographic order. That way, we compute the homotopy type $\check{\text{Cech}}\left(\mathcal{F}_{\leq a}^m; \frac{r}{2} = 1\right)$ by relating it to consecutively adding vertices of $\mathbb{I}^m$, the unit hypercube of m th dimension, to point 0. Further, for higher r , we propose a conjecture regarding contractibility of a subcomplex implying a formula for the exact number of wedges of $\mathbb{S}^3$'s and $\mathbb{S}^4$'s in the Čech complex.

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1 Introduction

Let $\mathcal{F}_n^m = \{A \subset [m] : |A| = n\}$. We equip this space with the symmetric difference metric d ; $d(A, B) = |(A \setminus B) \cup (B \setminus A)|$. For S a finite subset of a metric space X , and a scalar $r \geq 0$, the Čech complex $\check{\text{Cech}}(S; r)$ is an abstract simplicial complex whose vertex set is S and $\sigma \subset S$ is a simplex if and only if $\bigcap_{x \in \sigma} B\left[x, \frac{r}{2}\right] \neq \emptyset$ where $B[x, \frac{r}{2}]$ is a closed ball of radius $\frac{r}{2}$ around x . We know that a subset of a simplex is also a simplex.

If we have two subsets $a, b \subset [m]$, i.e., $a, b \in P([m])$, we arrange them in the ascending order in the following way:

- If $|a| > |b|$, we also have $a \succ b$, where $|a|$ is the cardinality of a .
- If $|a| = |b|$, then let $a = \{i_1, i_2, \dots, i_{|a|}\}$ and $b = \{j_1, j_2, \dots, j_{|a|}\}$. Let k be the smallest index such that $i_k \neq j_k$. Then $a < b$ accordingly as $i_k < j_k$ etc.

We can see this in an alternative manner. See each $a = \{i_1, \dots, i_{|a|}\} \in P([m])$ as an element in the m -dimensional unit hypercube $\mathbb{I}^m$, with each i_k th coordinate 1 for $k \in \{1, \dots, |a|\}$ and all other coordinates 0. In this scenario, our ordering is equivalent to:

- If $|a - 0| > |b - 0|$, we also have $a \succ b$, where $|a - 0|$ is the Hamming distance of a from 0, i.e., $|a - 0| = \sum_{i=1}^m |a_i - 0|$.
- Between points with the same distance from $(0, \dots, 0)$, we follow the reverse of lexicographic order.

We call this the *Cardinality Reverse-Lexicographic (CRL) order*. Also, observe that with this notation, if we drop the subsequent 0's after the last non-zero coordinates, we can view our point in $\mathbb{I}^m$ for a lesser m (with easier visualization), but that does not affect our calculations. Because coordinates in higher dimension don't bother us here, and also we can carry forward our results to higher dimension without any change, just by reverting to coordinates with the latter entries "0".

We wish to compute $\check{\text{Cech}}(\mathcal{F}_n^m; r)$, $r > 0$. When we are looking at $\mathcal{N}(\mathcal{F}_2^m; 2)$, the facets are $N[\{x, y\}] = \{\{i, x\} : i \in [m] \setminus x\} \cup \{\{j, y\} : j \in [m] \setminus y\}$. In other words, $\sigma \in E(K_m)$ forms a simplex is $\exists x, y$ such that $\forall e \subset \sigma$ either $x \in e$ or $y \in e$ or both. Given $H \in E(G)$ we say that a subset $A \subseteq V(G)$ covers H is every edge in H intersects A . Covering number of H is defined as $\min_{A \subseteq G} \{\#A : A \text{ is a cover of } H\}$. Also, $\sigma \in \mathcal{N}(\mathcal{F}_2^m; 2)$ if and only if $\text{cov}(\sigma) \leq 2$. In [J, Section 26.6], $\text{Cov}_{m,2}$ corresponds to our $\mathcal{N}(\mathcal{F}_2^m; 2)$, which should be homotopic to a wedge of $\mathbb{S}^3$'s and $\mathbb{S}^4$'s. However, the exact number of wedge components are not known, and is worth investigation.

Feng and Nukala's paper [FN] computes homotopy types for Vietoris-Rips complexes of such spaces. In this paper, we exhaustibly describe the open balls in $\check{\text{Cech}}(\mathcal{F}_n^m; r)$ and in the subsequent section, using computations for smaller n , inductively prove a formula for the homotopy types of $\check{\text{Cech}}(\mathcal{F}_n^m; r = 2)$ and $\check{\text{Cech}}(\mathcal{F}_{\leq a}^m; r = 2)$ for some $a \in \mathcal{F}_n^m$. In the last section, we talk about future directions, and show an approach to compute the homotopy types for general r , calculating the exact number of $\mathbb{S}^3$'s and $\mathbb{S}^4$'s in the wedge.

2 Preliminaries

2.1 Simplicial complexes

A *simplicial complex* L on a finite set V is a family of non-empty subsets of V such that if $\sigma \in L$ and $\tau \subseteq \sigma$ is non-empty, then $\tau \in L$.

Definition 2.1 (Nerve complex). Let $\mathcal{U} = \{U_i\}_{i \in \{1, \dots, k\}}$ be a collection of subsets of a metric space X . The nerve of $\mathcal{U}$, $\mathcal{N}(\mathcal{U})$ is a simplicial complex such that

- the vertex set is $\mathcal{U}$;
- a subset $\sigma \subseteq \mathcal{U}$ is a simplex if and only if $\bigcap_{U \in \sigma} U \neq \emptyset$.

A Čech complex is the nerve of the corresponding collection of r -balls, i.e., $\check{\text{Cech}}(S; r) = \mathcal{N}(\{B(s, r)\}_{s \in S})$.

2.2 The link-deletion method

For any vertex v in a complex K , deletion $K \setminus v$ denotes the induced complex on the vertex set $K^{(0)} \setminus \{v\}$. The link of v in K , $\text{lk}_K(v) = \{\sigma : \sigma \cup \{v\} \in K \text{ and } v \notin \sigma\}$.

We have the following lemma as an useful computational tool:

Lemma 2.2. [?] If K is a simplicial complex and for $K = K_1 \cup K_2$, the inclusion maps $i_j : K_1 \cap K_2 \hookrightarrow K_j$ are both null-homotopic. Then $K \simeq K_1 \vee K_2 \vee \Sigma(K_1 \cap K_2)$.

As a corollary, we have,

Lemma 2.3. [AA, Lemma 2] Let $v \in K$ be a vertex and the inclusion $\text{lk}_K(v) \hookrightarrow K$ is null-homotopic. Then, $K \simeq K \setminus v \vee \Sigma(\text{lk}_K(v))$.

2.3 Homotopy type of a complete graph

Definition 2.4. A complete graph is a simple undirected graph in which every pair of distinct vertices has a unique edge between them. The complete graph for n vertices, $n \in \mathbb{N}$, is denoted by K_n .

Any connected topological graph is homotopic to a wedge sum of $\mathbb{S}^1$'s, the number of which is given by the formula $\# \text{edges} - \# \text{vertices} + 1$. So,

$$K_n \simeq \bigvee_{|E(K_n)| - |V(K_n)| + 1} \mathbb{S}^1 = \bigvee_{\binom{m}{2} - m + 1} \mathbb{S}^1 = \bigvee_{\binom{m-1}{2}} \mathbb{S}^1.$$

Using this result, we shall show in Eq (2) that the homotopy type of $n + 1$ pairwise point-intersecting n -simplices is a wedge of $\mathbb{S}^1$'s. We also use the Nerve Theorem.

Theorem 2.5. [Nerve theorem] Let F be a finite collection of closed, convex sets in the Euclidean space. Then the nerve of F and the union of the sets in F have the same homotopy type.

3 Observations about Čech $(\mathcal{F}_n^m; r)$

3.1 The open balls

3.1.1 The case $n = 1$

For $n = 1$, the elements of $\mathcal{F}_1^m$ are singletons $\{x\}$ for $x \in [m]$. Therefore, $d(\{x\}, \{y\}) = 2$. We have that all singletons of singletons belong to $\check{\text{Cech}}(\mathcal{F}_1^m; r)$ since if $\sigma = \{\{x\}\}$, then $\bigcap_{v \in \sigma} B\left[v, \frac{r}{2}\right] \neq \emptyset$, since v has only one option $\{x\}$ and the intersection is trivially a singleton. For $0 \leq \frac{r}{2} < 2$ i.e., $0 \leq r < 4$, $B\left[v, \frac{r}{2}\right] = \{\{v\}\}$, while $B\left[v, \frac{r}{2} \geq 2\right] = \{\{i\} : i \in [m]\}$, and hence $\langle \{1\}, \dots, \{m\} \rangle$ is an $(m - 1)$ -simplex generated by all of the points.

3.1.2 The case $n = 2$

For $n = 2$, we first compute possible distances.

- $d(\{x, y\}, \{x, y\}) = 0$;
- $d(\{x, y\}, \{y, z\}) = 2$;
- $d(\{x, y\}, \{u, v\}) = 4$.

Let $v = (a, b)$ where $a < b \in [m]$. Then $B\left[v, \frac{r}{2} < 2\right) = \{v\}$. Also,

$$B\left[v, 4 > \frac{r}{2} \geq 2\right] = \{\{a, i\} \{j, b\} : a \neq i \in [m], b \neq j \in [m]\},$$

since at distance 0 lies the point $\{a, b\}$ itself at exactly distance 2 from v lie those points whose one entry is a and the other something else; and at distance 4 lies every other point in the space. Again, $B\left[v, \frac{r}{2} \geq 4\right]$ again consists of all the points in the space. For $\frac{r}{2} < 4$, a simplex would be generated by $\{\{a, i\} : a \neq i \in [m]\}$ so that the intersection will be non-empty.

3.1.3 The case $n = 3$

We proceed in a similar manner, with possible distances:

- $d(\{x, y, z\}, \{x, y, x\}) = 0$;
- $d(\{x, y, z\}, \{x, y, w\}) = 2$;
- $d(\{x, y, z\}, \{x, u, v\}) = 4$;
- $d(\{x, y, z\}, \{u, v, w\}) = 6$.

Let $V = (a, b, c)$ where distinct $a, b, c \in [m]$.

- $B\left[V, 0 \leq \frac{r}{2} < 2\right] = \{v\}$; hence each simplex is of the form $\sigma = \{\{V\}\}$ for $V \in \mathcal{F}_3^m$.
- $B\left(V, 2 \leq \frac{r}{2} < 4\right) = \{A \in \mathcal{F}_3^m : \exists B \in \mathcal{F}_2^m \ni B \subset V \text{ and } B \subset A\}$; hence each simplex is of the form $N^{(1)}[i_1, i_2, i_3] = \{A \in \mathcal{F}_3^m : \{i_1, i_2\} \subset A \text{ or } \{i_3, i_4\} \subset A\}$. Here $i_j \in [m]$ and $i_{j_1} \neq i_{j_2}$. In other words, $N^{(1)}[v] = \{A \in \mathcal{F}_3^m : \exists B \in \mathcal{F}_2^m \ni B \subset V \text{ and } B \subset A\}$.
- $B\left(V, 4 \leq \frac{r}{2} < 6\right) = \{A \in \mathcal{F}_3^m : \exists B \in \mathcal{F}_1^m \ni B \subset V \text{ and } B \subset A\}$; hence each simplex is of the form $N^{(2)}[i_1, i_2, i_3] = \{A \in \mathcal{F}_3^m : \{i_1\} \subset A \text{ or } \{i_2\} \subset A \text{ or } \{i_3\} \subset A\}$. Here $i_j \in [m]$ and $i_{j_1} \neq i_{j_2}$. Hence $N^{(2)}[V] = B\left[V, 4 \leq \frac{r}{2} < 6\right]$.
- $B\left[V, \frac{r}{2} \geq 6\right] = \mathcal{F}_3^m$. The whole space is a $(m^3 - 1)$ -simplex.

3.1.4 The generalization

In general, for $V \in \mathcal{F}_n^m$,

- $B\left[V, \frac{r}{2}\right] \ (0 \leq r < 4) = \{v\}$; hence each simplex is of the form $\sigma = \{\{V\}\}$ for $V \in \mathcal{F}_n^m$.

- For $4 \leq r < 8$, each simplex is of the form

$$N^{(1)}[V] = \{A \in \mathcal{F}_n^m : \exists \{i_1, i_2, \dots, i_{n-1}\} \subset V \ni \{i_1, i_2, \dots, i_{n-1}\} \subset A\},$$

where $V \in \mathcal{F}_n^m$.

- For $8 \leq r < 12$, each simplex is of the form

$$N^{(2)}[V] = \{A \in \mathcal{F}_n^m : \exists \{i_1, i_2, \dots, i_{n-2}\} \subset V \ni \{i_1, i_2, \dots, i_{n-2}\} \subset A\}.$$

$\vdots$

- $B(V, r \geq 2n) = \mathcal{F}_n^m$. the whole space is a $(m^n - 1)$ -simplex.

For $4k \leq r < 4(k+1)$, each simplex is of the form

$$N^{(k)}[V] = \{A \in \mathcal{F}_n^m : \exists \{i_1, i_2, \dots, i_{n-k}\} \subset V \ni \{i_1, i_2, \dots, i_{n-k}\} \subset A\},$$

where $i_j \in [m]$ and $i_{j_1} \neq i_{j_2}$.

4 Inductively, for $\mathcal{F}_2^m$

Clearly, G_1 consists of a single point $(0, 0, 0)$, hence Čech $\left(G_1, \frac{r}{2} = 1\right)$ is contractible. Similarly, G_2 and G_3 are contractible.



Figure 1: G_2

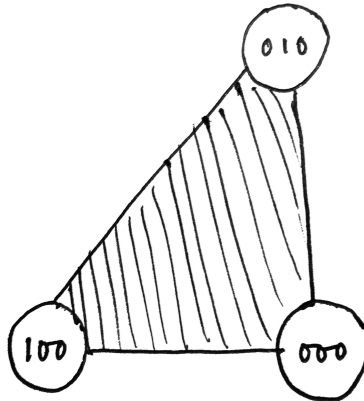


Figure 2: G_3

4.1 G_4

As for G_4 , we have 4 points: $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, and their neighbourhoods

- $N[(0, 0, 0)] = \{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$;
- $N[(1, 0, 0)] = \{(1, 0, 0), (0, 0, 0)\}$;
- $N[(0, 1, 0)] = \{(0, 1, 0), (0, 0, 0)\}$;
- $N[(0, 0, 1)] = \{(0, 0, 1), (0, 0, 0)\}$. So, G_4 is contractible.

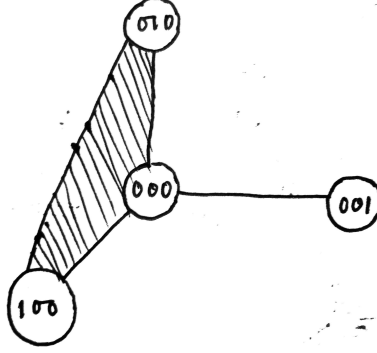


Figure 3: G_4

4.2 G_5

We have $\check{\text{Cech}}(G_5) = \langle N[(0, 0, 0)], N[(1, 0, 0)], N[(0, 1, 0)], N[(0, 0, 1)], N[(1, 1, 0)] \rangle$.

- $N[(0, 0, 0)] = \{(0, 0, 0), (0, 0, 1), (1, 0, 0), (0, 1, 0)\}$;
- $N[(1, 0, 0)] = \{(1, 0, 0), (0, 0, 0), (1, 1, 0)\}$;
- $N[(0, 1, 0)] = \{(0, 1, 0), (1, 1, 0), (0, 0, 0)\}$;
- $N[(0, 0, 1)] = \{(0, 0, 1), (0, 0, 0)\}$;
- $N[(1, 1, 0)] = \{(1, 1, 0), (0, 1, 0), (1, 0, 0)\}$.

Now, $\text{del}((1, 1, 0)) = \left\{ \sigma \in \check{\text{Cech}}(G_5) : (1, 1, 0) \notin \sigma \right\} = \check{\text{Cech}}(G_4) \simeq *$. Also, $\text{lk}((1, 1, 0)) = \langle N[v] : v \in N[(1, 1, 0)] \rangle = \Delta \simeq \mathbb{S}^1$. By the link-deletion method, $\check{\text{Cech}}(G_5) \simeq \mathbb{S}^2$.

For visualization, one can imagine filling in the discreteness, G_5 will have the structure of two tetrahedra, one hollow and one solid, attached at the surface $\{(0, 0, 0), (1, 0, 0), (0, 1, 0)\}$ [Fig 4].

This will be our general technique now on. We shall take the last-added point v , and in general,

$$\check{\text{Cech}}(G_k) \simeq \text{del}(v) \vee \Sigma \text{lk}_{\check{\text{Cech}}(G_k)}(v) = \check{\text{Cech}}(G_{k-1}) \vee \Sigma \text{lk}_{\check{\text{Cech}}(G_k)}(v). \quad (1)$$

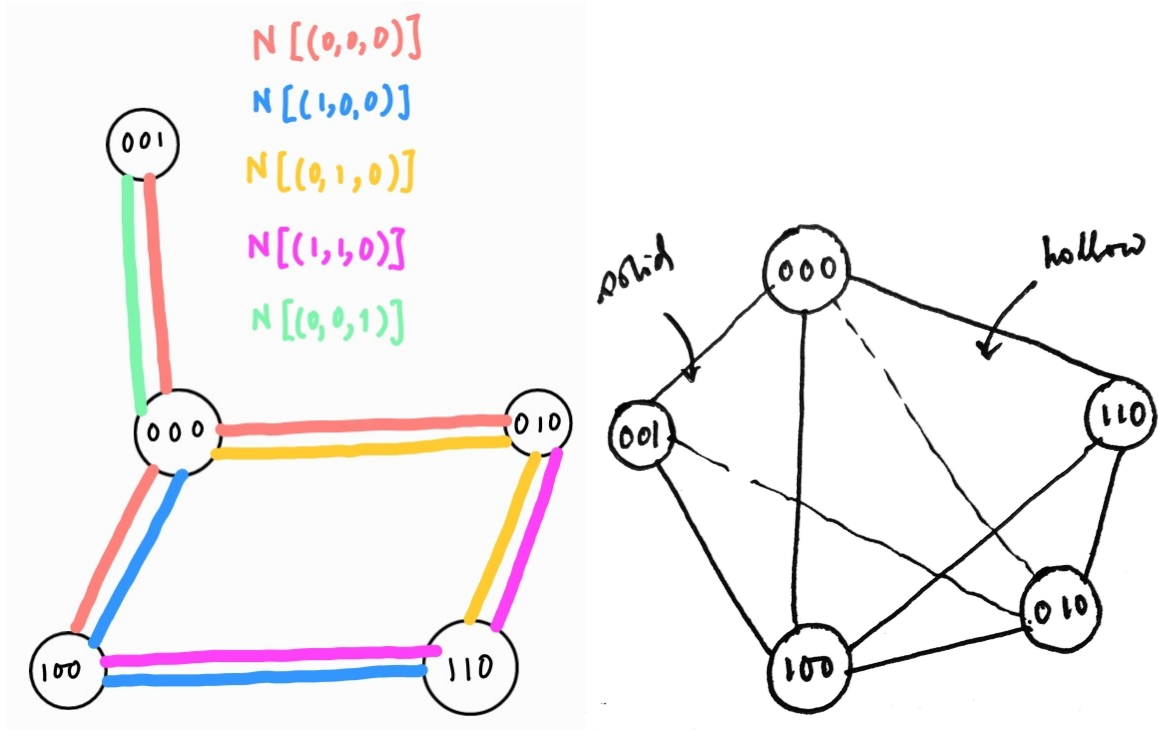


Figure 4: G_5

4.3 G_6

We have $\check{\text{Cech}}(G_6) = \langle N[(0,0,0)], N[(1,0,0)], N[(0,1,0)], N[(0,0,1)], N[(1,1,0)], N[(1,0,1)] \rangle$, where

- $N[(0,0,0)] = \{(0,0,0), (0,0,1), (1,0,0), (0,1,0)\}$;
- $N[(1,0,0)] = \{(1,0,0), (0,0,0), (1,1,0), (1,0,1)\}$;
- $N[(0,1,0)] = \{(0,1,0), (1,1,0), (0,0,0)\}$;
- $N[(0,0,1)] = \{(0,0,1), (0,0,0), (1,0,1)\}$;
- $N[(1,1,0)] = \{(1,1,0), (0,1,0), (1,0,0)\}$;
- $N[(1,0,1)] = \{(1,0,0), (0,0,1), (1,0,1)\}$.

In Figure 5, tetrahedra $\langle (0,0,0), (0,0,1), (1,0,0), (0,1,0) \rangle$, $\langle (1,0,0), (0,0,0), (1,1,0), (1,0,1) \rangle$ are solid, and the other two are hollow. Then, $\text{del}(1,0,1) \simeq \check{\text{Cech}}(G_5) \simeq \mathbb{S}^2$ and $\text{lk}(1,0,1) = \text{hollow } \langle (0,0,1), (1,0,0), (0,0,0) \rangle \simeq \mathbb{S}^1$. Then $\text{lk}(1,0,1)$ is contractible inside $\text{del}(1,0,1)$. Thus, $\check{\text{Cech}}(G_6) \simeq \mathbb{S}^2 \vee \mathbb{S}^2$. On similar calculations, we get $G_7 \simeq \bigvee_4 \mathbb{S}^2$.

4.4 For sets of cardinality 2

Alternatively, we have the following lemma:

Lemma 4.1. *By moving ahead with one more vertex α in the CRL order, as long as $|\alpha| = 2$, we wedge our previous Čech with an $\mathbb{S}^2$.*

Proof. Let α be the last added vertex. Supposing $\check{\text{Cech}}[\alpha] = \langle \alpha, \alpha_1, \alpha_2 \rangle$, we have, since $\langle \alpha_1, \alpha_2, 0 \rangle$ goes inside $\check{\text{Cech}}[0]$, three facets containing α :

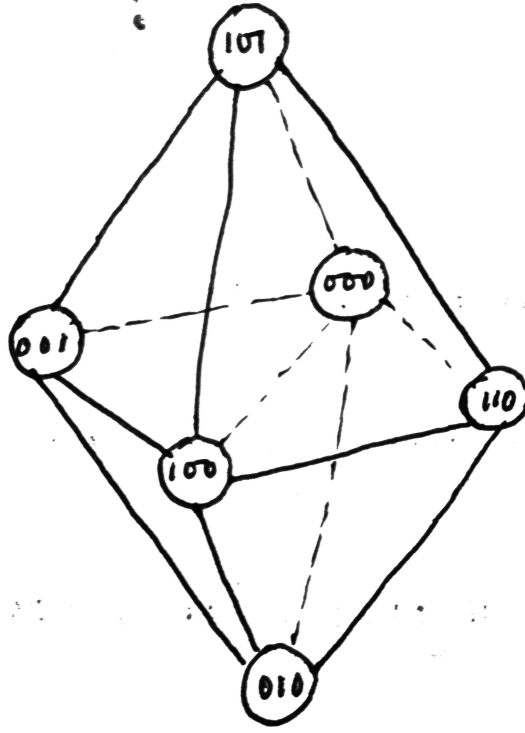


Figure 5: G_6

- $\check{\text{Cech}}[\alpha_1] = \langle \alpha_1, \alpha, 0 \rangle$,
- $\check{\text{Cech}}[\alpha_2] = \langle \alpha_2, \alpha, 0 \rangle$,
- $\check{\text{Cech}}[\alpha]$.

We know, $\text{lk}(\alpha) = \left\{ \tau \in \check{\text{Cech}}(G_{\text{no. } \alpha}) \mid \alpha \notin \tau, \tau \cup \alpha \in \check{\text{Cech}}(G_{\text{no. } \alpha}) \right\}$, where no. α is the sequential numbering of graph when α is added. Removing α from the 3 new facets we get 3 lines: $\overline{\alpha_1 0}, \overline{\alpha_2 0}, \overline{\alpha_1 \alpha_2}$. Of course this is not a filled-in triangle, since then $\langle \alpha_1, \alpha_2, 0 \rangle \subset \text{lk}(\alpha) \Rightarrow \langle \alpha_1, \alpha_2, 0, \alpha \rangle \subset \check{\text{Cech}}(G_{\text{no. } \alpha})$, a contradiction. This $\text{lk}(\alpha)$ is contractible in $\check{\text{Cech}}(G_{\text{no. } \alpha-1})$, which is, by induction hypothesis, wedge of S^2 's. Thus, by induction, using Eq (1), we have $\check{\text{Cech}}(G_{\text{no. } \alpha}) \simeq \bigvee_{\text{no. } \alpha-4} S^2$. $\square$

Thus, $\check{\text{Cech}}(G_{\text{no. } \alpha}) \simeq \bigvee_{\binom{m}{2}} S^2$ when $\alpha = \max F_2^m$.

4.5 Generalization

Let us move to the general case. We add a point $x \in \mathbb{I}_m$, and x has n non-zero coordinates. If there are any previous points with n non-zero coordinates included in the complex, none are “adjacent” to x . So, basically, we get $(n+1)$ new facets:

- $N[x] = \langle x, x_1, x_2, \dots, x_n \rangle$;
- $N[x_1] = \langle x, x_1 x_1^{(1)}, x_1^{(2)}, \dots, x_1^{(n-1)} \rangle$;
- $N[x_2] = \langle x, x_2, x_2^{(1)}, x_2^{(2)}, \dots, x_2^{(n-1)} \rangle$;

⋮

- $N[x_n] = \langle x, x_n, x_n^{(1)}, x_n^{(2)}, \dots, x_n^{(n-1)} \rangle,$

where $N[x]$ is the simplex around x , x_i is the face of x with the i th non-zero-coordinate made 0, $x_i^{(j)}$ is the face of x_i with the j th non-zero-coordinate made 0. We have,

$$\text{lk}_{\check{\text{Cech}}(\text{no. } x)}(x) = \left\{ \langle x_1, x_2, \dots, x_n \rangle, \langle x_1, x_1^{(1)}, x_1^{(2)}, \dots, x_1^{(n-1)} \rangle, \langle x_2, x_2^{(1)}, x_2^{(2)}, \dots, x_2^{(n-1)} \rangle, \dots, \langle x_n, x_n^{(1)}, x_n^{(2)}, \dots, x_n^{(n-1)} \rangle \right\}.$$

Transliteration to our “subset of $[m]$ ” notation,

$$\begin{aligned} \text{lk}_{\check{\text{Cech}}(\text{no. } x)}(x) = & \left\{ \langle x \setminus \{i_1\}, x \setminus \{i_2\}, \dots, x \setminus \{i_n\} \rangle, \right. \\ & \langle x \setminus \{i_1\}, x \setminus \{i_1, i_2\}, x \setminus \{i_1, i_3\}, \dots, x \setminus \{i_1, i_n\} \rangle, \dots, \\ & \left. \langle x \setminus \{i_n, i_1\}, x \setminus \{i_n, i_2\}, \dots, x \setminus \{i_n, i_{n-1}\} \rangle \right\}. \end{aligned}$$

These are n -simplices, $(n + 1)$ in number, any 2 of which have exactly 1 point in common, and no two the same one. Irrespective of the dimension of the simplices, by the Nerve lemma, since all intersections are contractible, $\text{lk}_{\check{\text{Cech}}(G_{\text{no. } x})}(x)$ is homotopic to the union of the simplices, which in turn is homotopic to K_{n+1} , the complete graph with $n + 1$ vertices. The homotopy type is then $\bigvee_{\binom{n}{2}} \mathbb{S}^1$. Since

this is contractible inside $\text{del}_{\check{\text{Cech}}(G_{\text{no. } x})}(x)$, a wedge of $\mathbb{S}^2$'s, using Eq 1,

$$\begin{aligned} \check{\text{Cech}}(G_{\text{no. } x}) & \simeq \check{\text{Cech}}(G_{\text{no. } x-1}) \vee \Sigma \text{lk}_{\check{\text{Cech}}(G_{\text{no. } x})}(x) \\ & \simeq \check{\text{Cech}}(G_{\text{no. } x-1}) \vee \Sigma \bigvee_{\binom{n}{2}} \mathbb{S}^1 \\ & = \check{\text{Cech}}(G_{\text{no. } x-1}) \vee \bigvee_{\binom{n}{2}} \mathbb{S}^2 \end{aligned} \quad (2)$$

Using induction, for any $m, n \in \mathbb{N}$ and $a \in \mathcal{F}_n^m$,

$$\begin{aligned} \check{\text{Cech}}(\mathcal{F}_{\leq a}^m) & = \bigvee_{\binom{m}{2}} \binom{2}{2} \mathbb{S}^2 \vee \bigvee_{\binom{m}{3} \binom{3}{2}} \mathbb{S}^2 \vee \dots \vee \bigvee_{\binom{m}{n-1} \binom{n-1}{2}} \mathbb{S}^2 \vee \bigvee_{\#\{b \in \mathcal{F}_n^m \mid b \leq a\}} \mathbb{S}^2 \\ & = \bigvee_{\sum_{i=2}^{n-1} \binom{m}{i} \binom{i}{2} + \#\{b \in \mathcal{F}_n^m \mid b \leq a\}} \mathbb{S}^2. \end{aligned} \quad (3)$$

5 For general r

5.1 Calculations using Python program

5.2 For any $m \in \mathbb{N}, n = 2, 3$

Lemma 5.1. Consider $A = \{\{i_1 < i_2 < i_m\} : \{i_1 < i_2\} \in \mathcal{F}_2^{m-1}\} \cong \mathcal{F}_2^{m-1}$. Clearly, $A \subset \mathcal{F}_3^m$. Then $\pi : \mathcal{F}_3^m \rightarrow A; \{i_1 < i_2 < i_m\} \mapsto \{i_1 < i_2 < m\}$ is a contraction.

Proof. If $A \subset X$ is closed, then $f : X \rightarrow A$ is a contraction if $f|_A = \text{id}_A$ and $d(f(x), f(y)) \leq d(x, y)$. Let $x = (i_1, i_2, i_3), y = (j_1, j_2, j_3)$. Then

$$d(f(x), f(y)) = d(\{i_1, i_2, m\}, \{j_1, j_2, m\}) = \begin{cases} 4 & \text{if } \{i_1, i_2\} \cap \{j_1, j_2\} = \emptyset; \\ 2 & \text{if } \{i_1, i_2\} \cap \{j_1, j_2\} = 1; \\ 0 & \text{if } \{i_1, i_2\} = \{j_1, j_2\} \end{cases}$$

and

$$d(x, y) = d(\{i_1, i_2, i_3\}, \{j_1, j_2, j_3\}) = \begin{cases} 6 & \text{if } \{i_1, i_2\} \cap \{j_1, j_2\} = \emptyset \text{ and } j_1 \neq j_2; \\ 4 & \text{if } \{i_1, i_2\} \cap \{j_1, j_2\} = 1 \text{ or if } j_1 = j_2 \text{ but not both;} \\ 2 & \text{if } \{i_1, i_2\} = \{j_1, j_2\} \text{ or if } \{i_1, i_2\} \cap \{j_1, j_2\} = 1, i_3 = j_3; \\ 0 & \text{if } \{i_1, i_2\} = \{j_1, j_2\} \text{ and } i_3 = j_3. \end{cases}$$

In any case, $d(f(x), f(y)) \leq d(x, y)$. □

Then, by CITE, $\mathcal{H}_q(\check{\text{Cech}}(\mathcal{F}_2^{m-1}; r)) \hookrightarrow \mathcal{H}_q(\check{\text{Cech}}(\mathcal{F}_3^m; r))$ is an (injective) inclusion. Let $\Delta_m \approx \check{\text{Cech}}(\mathcal{F}_2^m; 4 \leq r < 8) = K_1 \cup K_2$ where $\Delta \geq K_1 = \langle \sigma_{i,m} : i \in [m-1] \rangle$, $K_2 = \langle \sigma_{x,y} : 1 \notin \{x, y\} \rangle = \langle \Delta_m \setminus \{ \sigma_{i,m} : i \in [m-1] \} \rangle$. Here $\sigma_{x,y}$ is a facet $B[\{x, y\}; 2 \leq \frac{r}{2} < 8]$.

We the following homotopy types using computer:

Now, we propose the following conjecture:

Conjecture 5.2. *The inclusion $i_2 : K_1 \cap K_2 \hookrightarrow K_2$ is null-homotopic, i.e., $K_1 \cap K_2$ is contractible inside K_2 .*

Note that K_1 is contractible since vertex $\{1, m\}$ appears in every facet, and K_1 is a cone with apex $\{1, m\}$. For K_2 , the initial set is $[m-1]$ to choose the elements in the vertices from, so it is like Δ_{m-1} with some additional 2-distance-edges, e.g., there are edges from $\{1, m\}$ and $\{2, m\}$ to $\{1, 2\}$ inside K_2 ; and $\{1, m\}$ and $\{2, m\}$ have an edge between them. Using Lemma 2.2, $K_2 \simeq \Delta_{m-1}$. If the Conjecture 5.2 is true, then

$$\begin{aligned} \Delta_m &\simeq K_1 \vee K_2 \vee \Sigma(K_1 \cap K_2) \\ &\simeq * \vee \Delta_{m-1} \vee \Sigma \left(\left(\bigvee_{\binom{m-1}{4} - \binom{m-2}{4}} \mathbb{S}^2 \right) \vee \left(\bigvee_{\binom{m}{4} - \binom{m-1}{4}} \mathbb{S}^3 \right) \right) \\ &\simeq \Delta_{m-1} \vee \left(\left(\bigvee_{\binom{m-2}{3}} \mathbb{S}^2 \right) \vee \left(\bigvee_{\binom{m-1}{3}} \mathbb{S}^3 \right) \right) \end{aligned}$$

This proves, using induction,

$$\check{\text{Cech}}(\mathcal{F}_2^m; 4 \leq r < 8) \simeq \left(\bigvee_{\binom{m-1}{4}} \mathbb{S}^3 \right) \vee \left(\bigvee_{\binom{m}{4}} \mathbb{S}^4 \right). \quad (4)$$

Further, we conjecture

Conjecture 5.3. *The following homotopy type:*

$$\check{\text{Cech}}(\mathcal{F}_3^m; 4 \leq r < 8) \simeq \left(\bigvee_{\gamma_m} \mathbb{S}^3 \right) \vee \left(\bigvee_{(m-4)\alpha_m} \mathbb{S}^4 \right) \quad (5)$$

where $\alpha_m = \binom{m}{4}$, $\gamma_m = (m-4)\alpha_m - \frac{3m+1}{4} \binom{m-2}{3}$.

A Appendix: Homotopy type of $n + 1$ pairwise point-intersecting n -simplices

This section has been entirely written by the free version of Claude AI. Let $\Delta_0, \dots, \Delta_n \subset \mathbb{R}^N$ be $n + 1$ many n -dimensional simplices such that

$$\Delta_i \cap \Delta_j \text{ is a single point for every } i \neq j.$$

Write $X = \bigcup_{i=0}^n \Delta_i$ and, for $i \neq j$, $p_{ij} := \Delta_i \cap \Delta_j$. For each point p occurring as some p_{ij} , set

$$S_p := \{i \in \{0, \dots, n\} : p \in \Delta_i\}, \quad s_p := |S_p| \geq 2.$$

Lemma A.1. (a) If $i \neq j$ both lie in S_p , then $p_{ij} = p$.

(b) The sets $\binom{S_p}{2}$, as p ranges over the distinct intersection points, partition the edge set of the complete graph K_{n+1} on $\{0, \dots, n\}$; in particular $\sum_p \binom{S_p}{2} = \binom{n+1}{2}$.

(c) For $p \neq p'$, $|S_p \cap S_{p'}| \leq 1$.

(d) Every nonempty intersection $\bigcap_{i \in I} \Delta_i$ ($I \subseteq \{0, \dots, n\}$, $|I| \geq 2$) is a single point, equal to p_{ij} for any $i, j \in I$; consequently it is nonempty iff $I \subseteq S_p$ for some p .

Proof. (a) Since $i, j \in S_p$, $p \in \Delta_i \cap \Delta_j = \{p_{ij}\}$, so $p = p_{ij}$.

(b) Every pair $\{i, j\}$ determines p_{ij} and so lies in $\binom{S_{p_{ij}}}{2}$; by (a) it lies in no other $\binom{S_p}{2}$.

(c) If $i \neq j$ lay in $S_p \cap S_{p'}$, (a) would force $p = p_{ij} = p'$.

(d) For $i, j \in I$ distinct, convexity gives $\bigcap_{k \in I} \Delta_k \subseteq \Delta_i \cap \Delta_j = \{p_{ij}\}$, so the intersection is $\emptyset$ or $\{p_{ij}\}$; in the latter case $I \subseteq S_{p_{ij}}$ by definition of $S_{p_{ij}}$. $\square$

Theorem A.2. X is homotopy equivalent to a wedge of circles,

$$X \simeq \bigvee_b S^1, \quad b = \sum_p (s_p - 1) - n,$$

the sum ranging over the distinct intersection points p .

Proof. **Step 1 (Nerve reduction).** Each Δ_i is convex, hence contractible, and by Lemma A.1(d) every nonempty finite intersection of the Δ_i is a single point, hence contractible too. So $\{\Delta_i\}_{i=0}^n$ is a good cover of X , and the Nerve Lemma [H, Cor. 4G.3] gives

$$X \simeq \mathcal{N},$$

where $\mathcal{N}$ is the simplicial complex on vertex set $\{0, \dots, n\}$ with faces I such that $\bigcap_{i \in I} \Delta_i \neq \emptyset$. By Lemma ??(d),

$$\mathcal{N} = \bigcup_p \sigma_p,$$

where σ_p denotes the full simplex on vertex set S_p .

Step 2 (Collapsing each block to a tree). For each p fix a spanning tree τ_p of the complete graph on S_p (e.g., the star from any vertex of S_p); $\tau_p \subseteq \sigma_p$ has $s_p - 1$ edges. Both τ_p and σ_p are contractible, so the inclusion $\tau_p \hookrightarrow \sigma_p$ is a homotopy equivalence. By Lemma A.1(c), any face of σ_p of dimension ≥ 1 (any subset of S_p of size ≥ 2) is a face of no other σ_q , so the elementary collapses realizing $\sigma_p \searrow \tau_p$ (“ σ_p collapses onto τ_p ”) are simultaneously valid collapses of $\mathcal{N}$ itself, for every p at once. Performing them all gives a simplicial collapse

$$\mathcal{N} \searrow G := \bigcup_p \tau_p,$$

in particular a homotopy equivalence $\mathcal{N} \simeq G$.

Step 3 (Euler characteristic). G is a graph on the $n + 1$ vertices $\{0, \dots, n\}$ with $\sum_p (s_p - 1)$ edges (the τ_p share vertices but, by the same face argument as in Step 2, never edges). G is connected: given i, j , the tree $\tau_{p_{ij}}$ spans $S_{p_{ij}} \ni i, j$ and so joins them by a path. A connected finite graph is homotopy equivalent to a wedge of $b = E - V + 1$ circles [H, 1.A], so

$$b = \sum_p (s_p - 1) - (n + 1) + 1 = \sum_p (s_p - 1) - n.$$

Combining the three steps, $X \simeq \mathcal{N} \simeq G \simeq \bigvee_b S^1$. □

Corollary A.3 (Generic position). *If the $\binom{n+1}{2}$ points p_{ij} are pairwise distinct (no three of the Δ_i share a common point), then every $s_p = 2$, so $\mathcal{N} = K_{n+1}$ already after Step 1 of the proof of Theorem A.2 (no collapsing is needed, since every block is already an edge), and*

$$X \simeq K_{n+1} \simeq \bigvee_{\binom{n}{2}} S^1.$$

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